

Game Development 1 – Fall 2025

Assignment 4: Capstone Design Project (30%)

Due date: Phase 1: November 17th, 2025 (at the latest – ASAP is preferable)

Phase 2: November 29th, 2025 at 11:59 pm

Phase 3: December 1st, in class.

Overview of assignment:

Building on a previous project you have created, refine the gameplay design so that the project is ready to show off in a portfolio.

Choose a project to use as a starting point:

1. A *Game Development 1* assignment such as the Board Game or Card Game

or

2. A prototype or assignment from a different course (e.g. *Programming*, *GS*, *AM3 flight model* etc.)

You should focus on gameplay mechanics, game balance, and creating interesting and engaging challenges for the player to face.

Adding feedback and visuals can also help to communicate the gameplay and design. See the Examples of Design Projects for more detailed options.

Phases Overview:

Phase 1: Short proposal of your capstone design project.

Due: ASAP but suggested by Nov 17th at the latest

Grading: Ungraded but should be approved prior to Phase 2

Phase 2: Execution of your proposed design project.

- a. **Summary Document - PDF File**
- b. **Rules/Controls Page**
- c. **Project Assets: Build & Project files**

Phase 3: Demonstration to the class of your GD1 project.

Learning outcomes

- Learn how to polish a rough gameplay demo into a completed state, ready for presentation.
- Demonstrate your understanding of game mechanics, challenges, and level design.
- Demonstrate your ability to refine, balance, and fine tune the gameplay of a game.
- Communicate your process and be able to offer rationale to back up your design decisions.

Examples of Design Projects:

Suggested Option:

- **Mechanic & Challenges:** Refine your core mechanic, identifying the central element that makes it engaging and interesting, and balancing and creating a sequence of challenges that teaches the player how to first use, then later master the mechanic. The difficulty of the challenges should start at an introductory level (Easy) and develop to an expert level (Hard).
 - *E.G. Rebalancing and refining the level design for a GS1 game.*

Alternative options:

- **Building a digital prototype of a physical game:** If you have a Card or Board game prototype that you want to develop further, you could create a digital version of it in a game engine of your choice. It doesn't have to be graphically polished but should clearly communicate the gameplay. Use the digital demo to further identify and balance the gameplay.
- **Building a polished physical prototype:** If you want to work with physical materials, you could create a refined prototype, which should include some graphical elements, and a "finished" prototype of a Card or Board game prototype.

Task Breakdown

Phase 1:

Create a short proposal document that outlines what you are hoping to accomplish. It is not for marks, but you should get instructor approval before starting on the main assignment (executed in Phase 2). It does not have to be extremely polished but should be clear and professional looking.

Goal: Identify the project you will start from, and the design goals you will complete in Phase 2.

Deliverables: A short document outlining your proposed project. This should be completed and discussed with the instructor prior to starting Phase 2.

Phase 2:

1. You should check with your instructor before proceeding too far here. Send it through Teams Chat if you can't wait for a class to get approval.
2. Implement your design changes to the project.
3. Do some playtesting to help refine your design changes and make sure the difficulty is appropriate, and the gameplay is clear.
4. Create your Summary Document and your Rules or Controls page.

Goal: Improve the game design of an existing project and prepare it for your portfolio.

Deliverable: The final, refined game demo with supporting documentation.

Submission Instructions & Deliverables:

Deliverables:

1. Summary Document - PDF File

- This should be a 1-page, polished document that would be appropriate for a portfolio and explains what the project is, how it works, and highlights what you did in the project. Indicate clearly which parts of the game you designed, refined, or improved
- Back up your design reasoning and decisions with design principles from class for full marks.
- Each person should create and submit this individually

Use these section headings and address each in your summary:

- **Game Overview**
- **Core Mechanic & Player Goal**
- **Design Improvements Made**
- **Balance & Challenge Progression**

2. Rules/Controls Page

- Include the controls (for digital projects) or game rules (for physical projects) as a separate page to help with playtesting. This should be polished and ready to display.
- This is meant to be simple but clear guide on how to play the game.

3. Project Assets:

Physical and Digital games will have different submission requirements:

a. For digital projects:

- **Unreal Project/Unity Project - Zip file**
 - The complete project. I'm not grading based on this but would like it for reference and archival purposes.
- **Exported playable build - Zip file**
 - The playable build. This is what will be primarily marked. Make sure the controls are

b. For physical projects:

- **The physical Board Game prototype.**
 - The completed prototype. You should be considering printing graphics and using more polished materials for this option.

Additional Notes on project naming:

Please hand in your Project files in a zip file with the name GD1_A4_FirstInitialLastInitial.zip and the build with GD1_A4_Build_FirstInitialLastInitial.zip. The Design Doc and Rules/Controls should be a PDF with a similar naming structure such as GD1_A4_Summary_DF.PDF

Grading Overview:

Assignment 4 – Grading Overview

Category	What is Evaluated
Core Mechanic – Clarity & Engagement	How clear, fun, and compelling the main gameplay mechanic is.
Core Mechanic – Balance & Challenge Progression	How well the gameplay is balanced and how difficulty ramps logically.
Design Improvements – Iteration & Refinement	Evidence of improving the game through iteration and playtesting.
Design Improvements – Reasoning & Justification	How well design decisions are explained and supported by course concepts.
Summary Document – Clarity & Structure	Organization and clarity of the 1-page summary; includes all required headings.
Rules/Controls Page	Clarity, completeness, and usability of rules or controls for playtesting.
Presentation & Polish – Visual/Audio Quality	Quality of visuals, audio, and overall polish in the game demo/prototype.
Presentation & Polish – Overall Usability	How intuitive and playable the game is; controls and instructions are easy to follow.

Group Work Guidelines:

You can work solo or in a group on a single project. If you choose to work in a group, **each student must have their own focus** and submit their own assignment. Even though the game is shared, your summary document should clearly explain **what you personally contributed** and why you made the design decisions you did.

Additional Group Submission Notes:

One person needs to submit the final build/project and rules/instructions documents. Submit it individually if you are concerned that they might forget.

Each Person should individually create and submit the summary document and proposal, since you are each doing something different with it.

Here are some examples of how you might divide responsibilities:

- **Digital Game - Mechanic & UI Split**
 - One student focuses on refining the core mechanics and balancing challenges.
 - Another student focuses on improving the user interface, menus, tutorials, and player feedback systems.
- **Digital Game - Systems & Polish Split**
 - One student refines gameplay systems, difficulty curves, and core interactions.
 - Another student adds visual polish, animations, sound effects, and game feel enhancements.
- **Physical Board/Card Game - Design & Production Split**
 - One student focuses on the rules, balance, and challenge progression.
 - Another student produces the physical prototype, including graphics, cards, and boards.
- **Shared Concept, Different Variations**
 - One student creates a “beginner-friendly” version with simpler mechanics.
 - Another creates an “advanced” version with new mechanics or expert challenges.

Key Principle: If you work in a group, the final game can be shared, but each student must **demonstrate meaningful individual contribution**. Your submission should clearly communicate your work and reasoning.